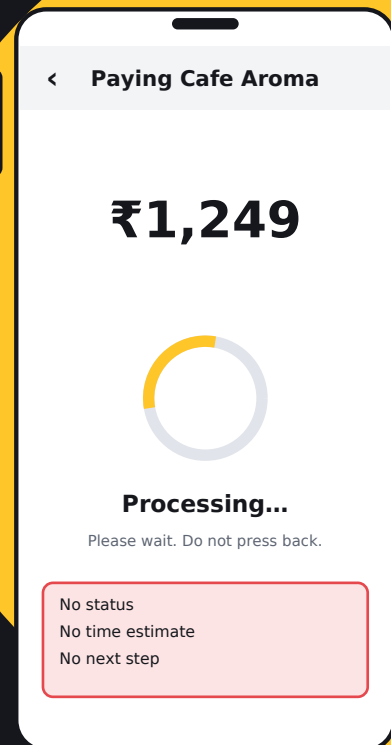


Payments stuck on “Processing”

Restoring trust in a credit-first UPI app by fixing what users see, do and get back when a payment hangs

Did it go through?

Money debited, no answer



Suchithra R

Product Management portfolio · September 2026



OVERVIEW

Contents

Executive summary 3

1. Context and problem framing 3

2. Users and moments that matter 5

3. Where reliability breaks 5

4. What users do when a payment hangs 7

5. Success metrics 8

6. Solutions 9

7. Prioritization and roadmap 11

8. Experiment plan 12

9. Risks and constraints 12

10. Assumptions and open questions 13

Sources 13

OVERVIEW

Executive summary

When a UPI payment hangs, the user does not know where their money is. This case study shows how to fix that.

1.2%

of payment attempts end in a pending state (assumed)

45%

of users retry within 5 minutes of a pending payment

3.5

payment tickets per 1,000 payments

–35%

target reduction in tickets

- **The problem.** In a credit-first UPI app, a payment that hangs on “Processing” leaves the user unsure whether money has moved. They retry, risk paying twice, contact support or switch apps.
- **Why it matters.** Daily UPI use is the habit that credit and cross-sell are built on. One confusing payment moment can cost more trust than many smooth ones earn, and it adds support cost.
- **The insight.** Most delay sits in bank and network layers the app does not own, and NPCI limits how often an app may check a payment’s status. So the winnable goal is not “never pending” but “**never confusing**”: say what is true, fast; prevent harmful retries; and resolve automatically.
- **The solutions.** (A) an honest pending state with a duplicate-payment guard, (B) automated recovery and refund tracking, (C) preflight reliability routing, and (D) a quick-win guard that pauses credit offers after a bad payment. Ranked with RICE and validated in A/B tests.
- **The targets.** Retry after pending from 45% to 20%, tickets per 1,000 payments from 3.5 to 2.3, and the 30-day retention gap for affected users from 7 points to 3.

SECTION 01

1

Context and problem framing

The product, the ecosystem it sits in, and the problem statement.

Product (assumed). A credit-first UPI payments app in the mould of super.money: UPI payments with cashback, a co-branded credit card, and credit or installment options on top of everyday payments.

Users. Everyday payers who scan QR codes, pay bills and rent, and a growing group who pay on credit. The app is a Third-Party Application Provider (TPAP): it does not run the rails, it depends on partner banks and the national UPI switch.

Landscape facts

Fact	Why it matters here	Source
super.money launched in August 2024 as a credit-first UPI app with up to 5% cashback	Credit cross-sell depends on payments being trusted	Credyfi
SplitStore, zero-interest installments inside the app, launched 6 August 2026	Credit is now a core revenue path, so payment trust has direct revenue stakes	TechTimes
UPI handled 22.35 billion transactions in April 2026	Even a small failure share is millions of stuck moments	productgrowth.in (NPCI data)
Technical declines fell from 8 to 10% (2016) to about 0.7 to 0.8% (2025); NPCI targets under 1%	The rails are good, so the remaining pain is the tail, and how it is communicated	D91 Labs; productgrowth.in
Failed UPI transactions must be auto-reversed in T+1 (fund transfers) or T+5 (merchant payments), with ₹100 a day owed for delays	The user is protected by rule, but the app rarely says so	RBI TAT framework via Ujjivan SFB, Zee News
In 2025 NPCI limited status checks: the first no sooner than 90 seconds after authentication, at most three in two hours	The app cannot simply poll harder; it must be smarter	D91 Labs
One user review describes QR payments stalling on “Processing” and needing two or three retries	Anecdotal, but it names the exact behaviour to fix	Google Play

Problem statement. When a payment hangs in a pending state, users cannot tell whether they have paid. They retry, risk duplicate payments, contact support or leave, which erodes trust in the app and weakens every credit cross-sell built on daily UPI use.

Scope

In scope	Out of scope
Pending and “Processing” experience, retry and duplicate handling, status communication, refund tracking, support deflection, pre-payment bank health signals, cross-sell suppression after a bad payment	Changing NPCI or partner-bank infrastructure, merchant-side apps, dispute adjudication, the fraud engine, new payment rails

Baseline snapshot (assumed, per 1,000 payment attempts)

OUTCOME OF 1,000 PAYMENT ATTEMPTS

Success		955
Business declines (wrong PIN, low balance)		25
Technical declines		8
Pending over 30 seconds		12

Pending is only about 1 in 83 attempts, but at UPI scale it touches a large number of people every day, and each of them has to decide what to do without a clear answer.

2

SECTION 02

Users and moments that matter

Who gets hurt most, and when the anxiety peaks.

Segment	Typical payment	What a pending state means to them	What they need
Everyday QR payers	Small, at a shop counter, in a queue	Social pressure; the merchant is waiting	A quick, believable proof they paid
Bill and rent payers	Medium, near a due date	Fear of late fees or a missed deadline	Clear status and a safe next step
High-value payers	Large, one-off	Fear of losing money	Reassurance backed by facts and a timeline
Credit-line users	Payment on credit or installments	Limit blocked; unsure if the repayment clock started	Status of both the payment and the limit

The five moments after “Processing” appears

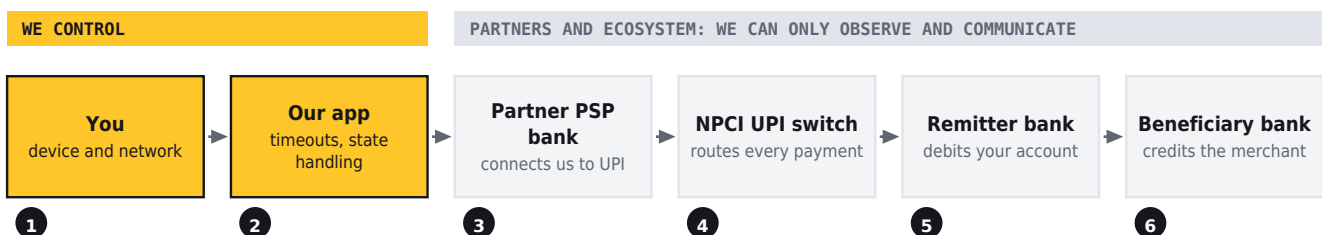
Moment	Timing	User state	What the app should do
M1 At the counter	0 to 30 seconds	Stressed, watching the screen	Show real status and what is safe to do
M2 Leaving	1 to 10 minutes	Tempted to retry or walk away	Warn against duplicate payments; offer proof
M3 Checking elsewhere	10 minutes to hours	Reading bank SMS, opening other apps	Push a clear final status as soon as it is known
M4 Next day	T+1	Waiting for money to return	Show refund progress and expected date
M5 After resolution	Days later	Deciding whether to trust the app again	Close the loop; hold back offers; repair trust

3

SECTION 03

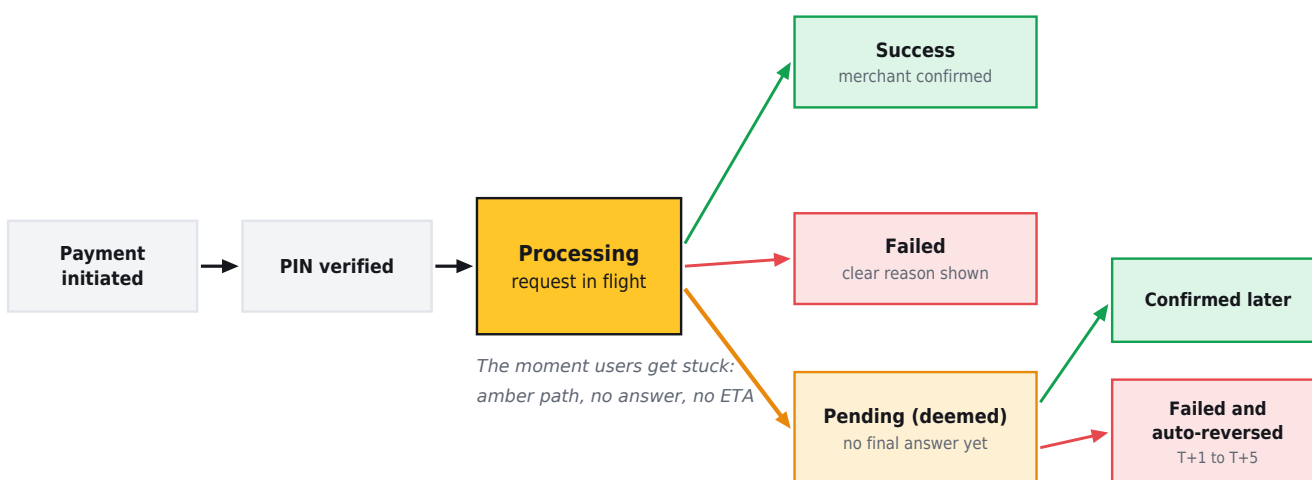
Where reliability breaks

A payment crosses six hands. We control the first two and must communicate for all six.



#	Layer	Typical failure	Who controls	What we can do
1	You: device and network	Connectivity drops mid-request; app backgrounded	Us (partly)	Idempotent request IDs; resume safely; clear offline messaging
2	Our app (TPAP)	Timeouts too aggressive or too lax; poor state handling	Us	Better timeouts and state machine; instrument every transition
3	Partner PSP bank	Slow responses, downtime	Partner	Track health by partner; escalate with data; route where possible
4	NPCI UPI switch	Load spikes and timeouts	Ecosystem	Respect API limits; use callbacks over polling
5	Remitter bank	Delays debiting or answering	Bank	Show per-bank health before payment; suggest another linked account
6	Beneficiary bank and merchant	Downtime leading to “deemed” status	Bank	Explain the state honestly; provide proof to the merchant

The payment state machine



Design implication. The pending path is where the user is left alone. It is not a bug we can remove, because it comes from bank and network conditions. It is a state we can design: say what we know, what is safe, and when they will hear next.

4

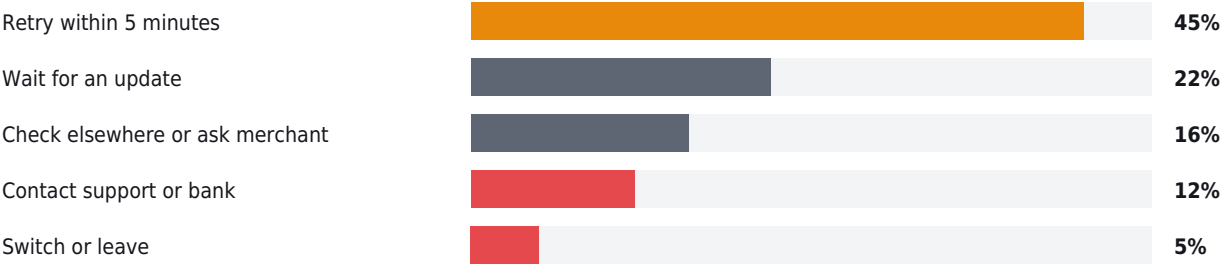
SECTION 04

What users do when a payment hangs

Seven reactions, why each happens, the risk it creates, and the design response.

Reaction	Why	Risk	Design response
Retry immediately	"Maybe it did not go through"	Duplicate payment	Duplicate guard; "do not pay again yet" message
Wait and stare	No information to act on	Anxiety, then abandonment	Live status with an expected time
Check bank SMS or another app	Wants a second source of truth	Confusion if sources disagree	Show the same status and reference the bank will show
Ask the merchant	Merchant is waiting for confirmation	Social pressure; disputes	Shareable, verifiable proof
Contact support	No self-serve answer	Cost and long wait	Self-serve status and refund tracker; prefilled tickets
Switch app or method	Trusts a rival more	Lost customer	Pre-payment health signals and alternatives
Abandon the purchase	Gave up under stress	Lost transaction	Fast, honest resolution and an easy "pay again"

WHAT PEOPLE DO AFTER A PENDING PAYMENT (ASSUMED SPLIT)



Assumed for illustration; the first analysis to run on real data. About 1 in 10 retries is assumed to end with both payments succeeding, which creates a duplicate debit. Pending-exposed users are assumed to churn at 12% in 30 days versus 5% for others.

5

SECTION 05

Success metrics

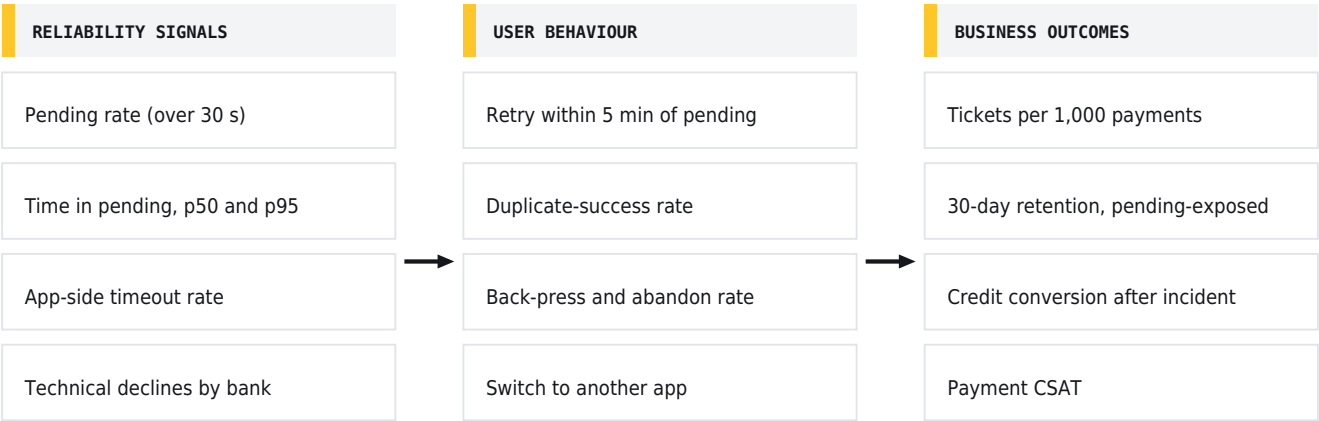
One north star, the signals that drive it, and the guardrails that protect users and the business.

North star: confident outcome rate. The share of payment attempts that end in a clear final state (success, or failure with a next step) within 30 seconds, with no retry caused by uncertainty and no support contact.

NORTH STAR

Confident outcome rate

95% → 97%



GUARDRAILS

False-reassurance rate · NPCI status-check limit breaches · Added checkout latency · Fraud via fake proofs

Metric definitions and targets

Metric	Definition	Baseline	Target
Confident outcome rate	Attempts with a clear final state in 30 s, no uncertainty retry, no contact	95%	97%
Pending rate	Attempts still pending after 30 s ÷ attempts	1.2%	0.9%
Retry within 5 min of pending	Users who re-attempt the same payment ÷ users seeing pending	45%	20%
Duplicate-success rate	Retries where both payments succeed ÷ retries	10%	3%
Tickets per 1,000 payments	Payment-related contacts ÷ payments × 1,000	3.5	2.3
30-day retention gap	Retention of pending-exposed users minus non-exposed	−7 pts	−3 pts
Credit conversion after incident	Credit activation within 14 days of a resolved pending event	4%	5%

Some pending events are outside our control (bank outages), so track “time to first clear message” and “time in pending” separately. Only the first is fully ours.

6

SECTION 06

Solutions

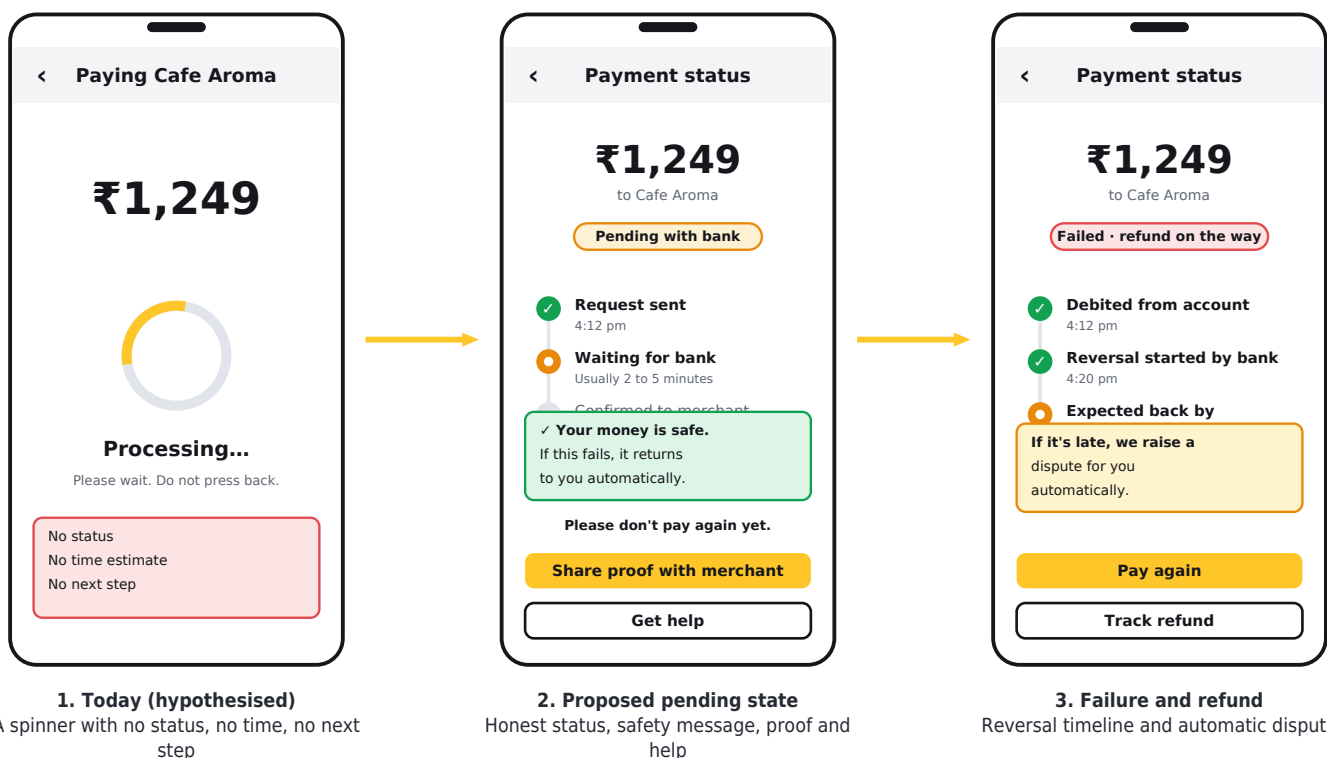
Four changes, from the honest pending state to a quick-win that protects revenue.

A. Honest pending state and duplicate guard

Problem it solves	Users see only “Processing” and guess, so they retry, doubt and leave.
What it is	After a short threshold, the app replaces the spinner with a status tracker : what happened, what we are waiting for, roughly how long, and whether the money is safe. It updates from callbacks instead of polling, in line with NPCI’s status-check limits.
Key behaviours	Plain language, no jargon. A “please do not pay again yet” prompt. A duplicate guard that warns when the same payee and amount are paid within ten minutes. A verifiable proof the user can show the merchant.
Where	Moments M1 to M3.
Metrics	Retry within 5 minutes, duplicate-success rate, back-press rate, contacts per pending event.
Risks	False reassurance if copy over-promises. Fake screenshots if proofs are not verifiable.

B. Automated recovery and refund tracking

Problem it solves	After a failure with a debit, users do not know when the money returns and must chase support.
What it is	Server-side reconciliation moves pending items to a final state and pushes the result. A refund tracker shows timeline and expected date based on the RBI turnaround times. If the timeline is missed, the app raises the dispute for the user and shows its progress, including any compensation the bank owes.
Credit-line payments	On failure, release the blocked limit at once and show it.
Where	Moments M3 to M5.
Metrics	Tickets per 1,000 payments, share of reversals inside the turnaround time, refund-related CSAT.
Risks	Dependence on partner data. Reading “fewer tickets” as success when problems are simply hidden.

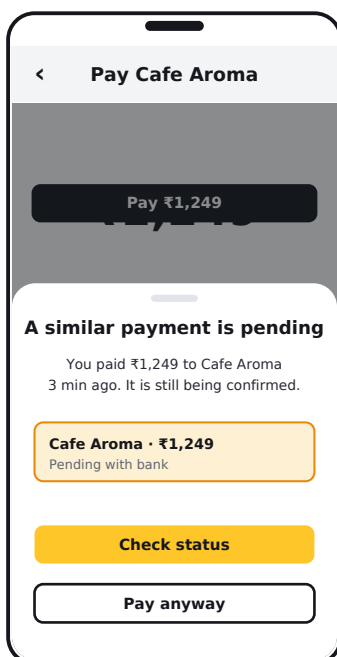


C. Preflight reliability routing

Problem it solves	Some payments are doomed before they start: a source bank is slow or degraded, and the user finds out only after waiting.
What it is	Track success rates by bank and time window. When a source is degraded, show a calm notice before payment and offer a healthier linked account or credit line. The user chooses; the app never switches funds silently.
How to start	In shadow mode first: log predictions against actual outcomes before showing anything.
Metrics	Pending rate, first-attempt success, warning precision, attempts abandoned after a warning.
Risks	Over-warning suppresses payments. Thresholds need tuning and partner buy-in.

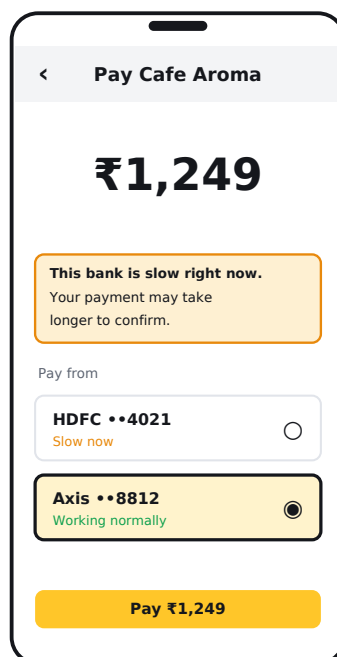
D. Quick win: cross-sell guard and trust repair

Problem it solves	Credit offers arriving right after a bad payment feel tone-deaf and lower conversion.
What it is	Suppress credit prompts and notifications for 48 hours after a pending or failed payment, or until it is resolved. After resolution, send a short “sorted” message, optionally with a small goodwill reward for a test cohort.
Metrics	Credit activation after incident, complaint rate, cost of goodwill.
Risks	Goodwill costs money and may attract gaming; test before scaling.



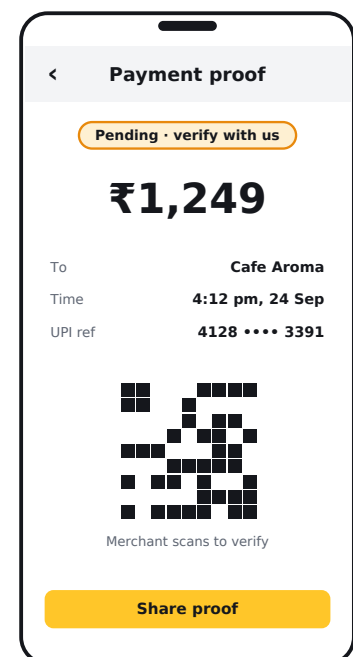
4. Duplicate guard

Stops a second payment while the first is pending



5. Preflight warning

A slow bank is flagged before paying, with an alternative



6. Proof for the merchant

A verifiable receipt that defuses counter pressure

Wireframes are low fidelity and show structure and behaviour only. Wording about refund timelines should be checked against current RBI and NPCI rules before launch.

7

SECTION 07

Prioritization and roadmap

RICE scores for every candidate, then a phased plan.

Reach is the share of active users touched per quarter. Impact: 0.5 low, 1 medium, 2 high. Effort is in person-months. $RICE = Reach \times Impact \times Confidence \div Effort$. All inputs are illustrative assumptions.

Candidate	Reach	Impact	Confidence	Effort	RICE
D. Cross-sell guard and trust repair	25	1	80%	1	20.0
A. Honest pending state and duplicate guard	25	2	80%	2.5	16.0
B. Automated recovery and refund tracking	25	2	60%	4	7.5
C. Preflight reliability routing	60	1	50%	5	6.0
E. Failure-prediction model	60	1	30%	6	3.0

Reading the scores

D scores highest because it is almost free, but it only protects revenue; A is the real fix and ships alongside it. B has higher impact over time but depends on partner data. C and E rest on bank-health signals we do not yet have, so we collect data first and keep E parked.

Expected impact

Solution	Main metric moved	Baseline → target
A. Honest pending state	Retry within 5 minutes; duplicate-success rate	45% → 20%; 10% → 3%
B. Recovery and tracking	Tickets per 1,000 payments	3.5 → 2.3
C. Preflight routing	Pending rate	1.2% → 0.9%
D. Cross-sell guard	Credit conversion after incident	4% → 5%
Combined	Confident outcome rate; retention gap	95% → 97%; -7 → -3 pts

Phased roadmap

Phase	Weeks	Scope	Exit criteria
0	0 to 3	Instrument every payment state change; classify tickets; mine reviews; set baselines	Trusted baseline for all metrics in section 5
1	3 to 10	A and D: honest pending state, duplicate guard, cross-sell guard	Retry rate down and guardrails flat
2	10 to 20	B: automated recovery, refund tracker, credit-limit release	Tickets per 1,000 payments trending to target
3	20 to 32	C in shadow mode, then live for a cohort; E stays parked	Warning precision high enough to go live

8

SECTION 08

Experiment plan

Every change ships behind a test with pre-agreed guardrails.

Test	Hypothesis	Design	Primary metric	Guardrails	Runtime
1. Pending state (A)	Honest status reduces harmful retries	50/50 among users who see pending	Retry within 5 minutes (45%, detect 5 pts: about 1,500 per arm)	False-reassurance rate, latency	2 weeks, then 30-day read
2. Cross-sell guard (D)	Pausing offers raises later conversion	50/50 among users after a resolved incident	14-day credit activation (4%, detect 1 pt: about 6,700 per arm)	Complaints, offer revenue	3 weeks
3. Recovery (B)	Tracking and auto-dispute cut contacts	Staggered rollout with a 10% holdout	Tickets per 1,000 payments	CSAT, resolved-in-TAT	6 weeks
4. Preflight (C)	Early warnings raise first-attempt success	Shadow mode 4 weeks, then 50/50 live	Confident outcome rate	Abandon after warning, false positives	4 + 4 weeks

- Test 1 also needs a 30-day retention read: baseline 88%, detect 2 points, about 4,100 users per arm.
- Randomize at user level. Reliability clusters at month-end, peak hours and bank outages, so run across full weeks and stratify by bank.
- Treat major outages as a separate segment; do not let them decide the result.
- Judge success on refund-adjusted, retained outcomes, not on tickets alone. Falling tickets can hide unresolved problems.

9

SECTION 09

Risks and constraints

Risk	Why it matters	Mitigation
False reassurance	Saying money is safe when it is not destroys trust	Base copy on the rules; state timelines, not guarantees; monitor mismatches
Fraud with fake proofs	Scammers can fake a “pending” screenshot	Verifiable receipts, short expiry, minimal fields
Over-warning	Too many bank warnings suppress payments	Tune thresholds; shadow mode first; measure abandonment
API limits	NPCI caps status checks	Callback-driven updates with backoff; never exceed limits
Partner dependency	Bank and PSP data drives C and B	Data-sharing agreements and escalation paths
Blame-shifting tone	Blaming banks looks defensive	Neutral, factual wording
Privacy	Proofs reveal payee and amount	Share only what is needed; expire links
Hidden problems	Ticket deflection can mask unresolved issues	Track resolution, CSAT and retention together
Compliance	Rules change	Verify current RBI and NPCI circulars before launch

10

SECTION 10

Assumptions and open questions

Assumption	How to confirm	Owner
Baselines: pending rate, retry rate, duplicate-success, tickets, retention	Payment event logs; ticket taxonomy; cohort analysis	Data
Pending distribution by bank, hour and month-end	Analytics by partner and time window	Data / Payments
User reactions after pending	Session replays; short surveys; support transcripts	Research
Partners can share health and reversal data	Discovery with PSP and banks	Partnerships
Credit-line payments follow the same states	Review credit integration	Engineering
Cost and effect of goodwill rewards	Small controlled pilot	Growth / Finance

OVERVIEW

Sources

Secondary sources used for context. Confirm regulatory points against the RBI and NPCI primary circulars.

Source	Used for	Link
Google Play: super.money reviews	User-reported stuck payments	play.google.com/store/apps/details?id=money.super.payments
Credyfi	super.money launch and positioning	credyfi.com/news/flipkart-backed-supermoney-unveils-upi-app
TechTimes, 7 Aug 2026	SplitStore launch and UPI fee legislation	techtimes.com/articles/323386/20260806
D91 Labs	Technical declines; NPCI 2025 operating limits	d91labs.substack.com/p/why-upi-success-rate-matters
productgrowth.in	NPCI decline metrics and UPI volume	productgrowth.in/insights/fintech/upi-payment-success-rates
Ujjivan Small Finance Bank	Reversal timelines (T+1, T+5) and compensation	ujjivansfb.bank.in/banking-blogs/banking-services/upi-payment-failed-refund-process-and-rules
Zee News, 5 Apr 2021	RBI harmonised turnaround-time framework	zeenews.india.com/personal-finance